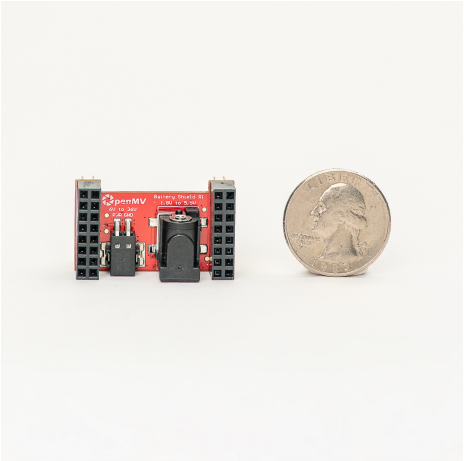
OpenMV® Battery Shield

Datasheet

SKU: OMV-BATTERY-SHIELD-V3

Description

The Battery Shield powers your OpenMV Cam from a 1.8 V to 5.5 V source connected via a standard 2.1 mm DC barrel jack — perfect for three alkaline batteries in series. It also features a wide voltage range 6\u201336 V input with reverse voltage and transient surge protection for industrial and automotive power. Includes a robust 0\u20136 V ADC input on P6 for measuring battery voltage. The shield powers your camera via the RAW pin, bypassing the onboard battery charger, and draws only microamps in operation — enabling years of battery life with cameras like the RT1062 in deep sleep (~30 \u00B5A).



Target Areas

Battery-powered deployments, remote sensing, field installations, alkaline battery operation, solar-powered systems, industrial monitoring, long-duration deployments.

Key Features

- **1.8–5.5 V DC input** — 2.1 mm DC barrel jack. Supports 2–3 alkaline batteries in series.
- **6–36 V wide-range input** — Reverse voltage and transient surge protection. Outputs 5.6 V on VIN via ideal OR'ing diode. Can be used simultaneously with DC input.
- **Battery voltage ADC** — Robust 0–6 V ADC input on P6 outputs 0–2.8 V for measuring battery voltage. Reverse voltage protected. Disconnectable.
- **Ultra-low quiescent** — Draws microamps in operation. Years of battery life possible in deep sleep.
- **RAW power path** — Powers camera via RAW pin, bypassing onboard battery charger.

Contents

1 Features

1.1 Low-Voltage Input

1.2 Wide-Range Input

1.3 Battery Monitoring

1.4 Power Path

1.5 Compatibility

2 General Info

3 Company Information

4 Reference Documentation

5 Revision History

1 Features

1.1 Low-Voltage Input

Component	Details
DC Barrel Jack	2.1 mm DC barrel jack. Accepts 1.8 V to 5.5 V. Perfect for 3 alkaline batteries in series (4.5 V nominal). 2 batteries also work but capacity is not fully usable. 4 batteries may exceed max voltage — proceed with caution.

1.2 Wide-Range Input

Component	Details
6–36 V Input	Robust to reverse voltage with transient surge protection. Outputs 5.6 V on VIN via ideal OR'ing diode for multi-shield stacking. Can be used simultaneously with the low-voltage input.

1.3 Battery Monitoring

Component	Details
ADC Input	Robust 0–6 V ADC input connected to DC barrel jack. Outputs 0–2.8 V on P6 for battery voltage measurement. Reverse voltage protected. Disconnectable.

1.4 Power Path

Component	Details
RAW Pin	1.8–5.5 V input directly powers camera via RAW pin, bypassing the onboard battery charger. Enables ultra-low quiescent current for deep sleep deployments.

1.5 Compatibility

Component	Details
Supported Cameras	All OpenMV Cam models. Optimized for cameras with deep sleep capability (e.g., OpenMV Cam RT1062 at ~30 µA deep sleep).

2 General Info

Dimension	Value
Weight	9 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/battery-shield
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.